

CS602 Computer Networks Architecture & Protocols

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Unit 1: Network Architectures & Physical Layer

OSI 7-Layer Model:

1. Physical (Bits, cables, modulation)
2. Data Link (Frames, MAC address, CSMA/CD, Error control)
3. Network (Packets, IP addressing, Routing)
4. Transport (Segments, End-to-end delivery, TCP/UDP port numbers)
5. Session (Dialog control, synchronization)
6. Presentation (Encoding, encryption, compression)
7. Application (User services, HTTP, DNS, SMTP, FTP).

Nyquist Maximum Bit Rate: $C = 2B \log_2(V)$ bps.

Shannon Channel Capacity with noise: $C = B \log_2(1 + S/N)$ bps.

Unit 2: Data Link Layer & Framing

Flow Control Mechanisms:

- Stop-and-Wait: Sender transmits one frame and waits for ACK. Efficiency = $1 / (1 + 2a)$, where $a = T_{prop} / T_{trans}$.
- Go-Back-N (Sliding Window): Sender window size $W_s = 2^k - 1$, Receiver window size $W_r = 1$.
- Selective Repeat: $W_s = W_r = 2^{(k-1)}$. Both sender and receiver maintain window buffers.

Unit 3: IPv4 Subnetting & Routing Algorithms

IPv4 Classes:

- Class A: 0.0.0.0 - 127.255.255.255 (Subnet mask 255.0.0.0)
- Class B: 128.0.0.0 - 191.255.255.255 (Subnet mask 255.255.0.0)
- Class C: 192.0.0.0 - 223.255.255.255 (Subnet mask 255.255.255.0)

Subnetting Math:

Borrowing n host bits gives 2^n subnets. Each subnet has $2^{(32 - \text{mask})} - 2$ usable host addresses.

Routing Protocols:

- Distance Vector (Bellman-Ford algorithm, e.g. RIP, subject to count-to-infinity problem).
- Link State (Dijkstra's shortest path algorithm, e.g. OSPF).

Unit 4: Transport Layer (TCP & UDP)

TCP 3-Way Handshake:

1. Client sends SYN (seq = x).
2. Server replies with SYN-ACK (seq = y , ack = $x + 1$).
3. Client sends ACK (seq = $x + 1$, ack = $y + 1$).

TCP Congestion Control States:

1. Slow Start: Congestion Window (cwnd) doubles every RTT until slow start threshold (ssthresh).
2. Congestion Avoidance: cwnd grows linearly (+1 MSS per RTT).
3. Fast Retransmit & Fast Recovery: Triggered by 3 duplicate ACKs.

